INTRODUCTION TO SYSTEMS ANALYSIS AND DESIGN:

AN AGILE, ITERATIVE APPROACH

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CHAPTER 3

Use Cases

Chapter 3

Introduction to Systems Analysis and Design: An Agile, Iterative Approach
6th Ed

Satzinger, Jackson & Burd

Chapter 3 Outline

- Use Cases and User Goals （用例和用户目标）
- Use Cases and Event Decomposition （用例和事件分解）
- Use Cases and CRUD （用例和 CRUD ）
- Use Cases in the Ridgeline Mountain Outfitters Case （ RMO 中的用例）
- Use Case Diagrams （用例图）

Learning Objectives

- Explain why identifying use cases is the key to defining functional requirements （为什么确定用例是定义功能性需求的关键）
- Describe the two techniques for identifying use cases （确定用例的两种技术）
- Apply the user goal technique to identify use cases （将用户目标技术运用于确定用例）
- Apply the event decomposition technique to identify use cases （将事件分解技术运用于确定用例）

Learning Objectives

- Apply the CRUD technique to validate and refine the list of use cases （将 CRUD 技术运用于确认和细化用例列表）
- Describe the notation and purpose for the use case diagram （描述用例图的符号和作用）
- Draw use case diagrams by actor and by subsystem （通过参与者和子系统画用例图）

开篇案例： Waiters on Call

- Waiters on Call 是在 2008 年由 Sue 和 Tom Bickford 开创的一项餐厅送餐服务。


开篇案例： Waiters on Call

- 当业务逐渐扩大时， Bickford 兄弟意识到他们需要一套专门的计算机系统来支持运营：
  - 记录顾客订餐需求
  - 餐厅查看顾客订单
  - 为司机派单
  - 司机更新空闲时间
  - 顾客修改订单需求
  - ……

Overview

- Chapter 2 provided an overview of systems analysis activities, functional and non-functional requirements, modeling, and information gathering techniques （第 2 章概述了系统分析活动、功能和非功能需求、建模和信息收集技术）
- This chapter focuses on identifying and modeling the key aspect of functional requirements– use cases （本章着重于识别和建模功能需求的关键方面——用例）

Overview

- In the RMO Tradeshow System from Chapter 1, some use cases are *Look up supplier*, *Enter/update product information*, *Enter/Update contact information* （在第一章的 RMO 展销系统中，一些用例是查找供应商、输入 / 更新产品信息、输入 / 更新联系信息）
- In this chapter’s opening case Waiters on Call, examples of use cases are *Record an order*, *Record delivery*, *Update an order*, *Sign in driver*, *Reconcile driver receipts, Produce end of day deposit slip,* and *Produce weekly sales reports* （在本章的开头案例“随叫随到的服务员”中，用例的示例有：记录订单、记录交货、更新订单、签到司机、核对司机收据、产生当天的存款）

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Use Cases

- Use case （用例） — an activity that the system performs, usually in response to a request by a user （系统执行的活动，通常是对用户请求的响应）
- Use cases define functional requirements(用例定义功能需求)
- Analysts decompose the system into a set of use cases (functional decomposition) （分析人员将系统分解为一组用例）

Use Cases

- Two techniques for Identifying use cases （识别用例的两种技术）
  - 1) User goal technique （用户目标技术）
  - 2) Event decomposition technique （事件分解技术）
- Name each use case using *Verb-Noun* *（动词-名词）*
  - 用例： do something

1) User Goal Technique （用户目标技术）

- Identify all of the potential categories of users of the system （识别系统的所有潜在用户类别）
- Interview and ask them to describe the tasks the computer can help them with （请他们描述本系统可以帮助他们完成的任务）
- Probe further to refine the tasks into specific user goals, "I need to *Ship items*, *Track a shipment*, *Create a return*" （进一步将任务细化为特定的用户目标，"我需要发送物品、跟踪发货、创建退货"）
- This technique is the most common in industry
- Simple and effective

1) User Goal Technique: Specific Steps

1. Identify all the potential users for the new system（定义新系统所有的潜在用户）
2. Classify the potential users in terms of their functional role (e.g., shipping, marketing, sales)（根据潜在用户的功能角色进行分类：运输、营销、销售）
3. Further classify potential users by organizational level (e.g., operational, management, executive)（通过组织的层次进一步将潜在用户分类：运营、管理、执行）

1) User Goal Technique Specific Steps

4. For each type of user, interview them to find a list of specific goals they will have when using the new system (current goals and innovative functions to add value) （采访用户收集系统目标）
4. Create a list of preliminary use cases organized by type of user （创建按用户类型组织的主要用例列表）
4. Look for duplicates with similar use case names and resolve inconsistencies （寻找相似的用例名称去掉重复用例）

1) User Goal Technique Specific Steps

7. Identify where different types of users need the same use cases （确定不同类型用户在哪里需要相同的用例）
7. Review the completed list with each type of user and then with interested stakeholders （与每种类型的利益相关者检查用例列表）

1) User Goal Technique

Some RMO CSMS Users and Goals

somebody do something

User	User goal and resulting use case
Potential customer	Search for item Fill shopping cart View product rating and comments
Marketing manager	Add/update product information Add/update promotion Produce sales history report
Shipping personnel	Ship items Track shipment Create item return

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2) Event Decomposition Technique （事件分解技术）

- More Comprehensive and Complete Technique （更全面和完整的技术）
  - Identify the events that occur to which the system must respond. （确定系统必须响应的事件）
  - For each event, name a use case (verb-noun) that describes what the system does when the event occurs （对于每个事件，命名一个用例：动词 - 名词）
- Event （事件） – something that occurs at a specific time and place, can be described, and should be remembered by the system （在特定时间和地点发生的事情）

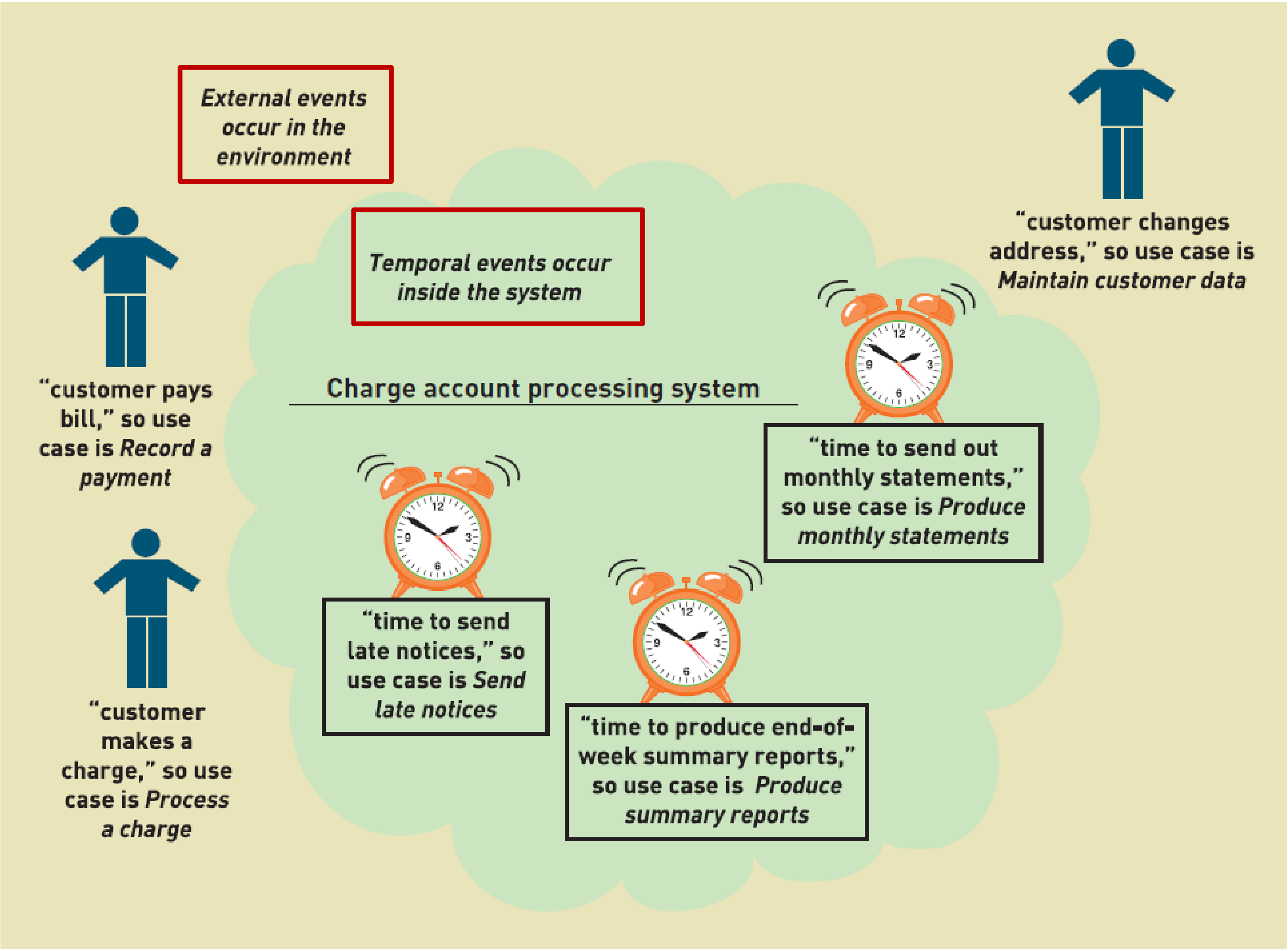
2) Event Decomposition Technique

- 用户提出了三个需求，哪一个是合适的用例?
  - A: 在表格中输入一个顾客姓名
  - B：增加新顾客信息
  - C：增加新顾客、删除顾客、跟踪预期付款的顾客、联系以前的顾客
- 用来确定用例的合适细节级别是基本业务流程（ Elementary Business Process, EBP ）。
  - 基本业务流程是由一个人在特定地点为了响应交易事件所执行的一项任务，它增加了可衡量的业务价值，保持了系统与数据的稳定和一致。

Types of Events

- External Event：外部事件
  - an event that occurs outside the system, usually initiated by an external agent or actor
- Temporal Event：时间事件
  - an event that occurs as a result of reaching a point in time
- State Event：状态事件（内部事件）
  - an event that occurs when something happens inside the system that triggers some process
  - reorder point is reached for inventory item

Events and Use Cases



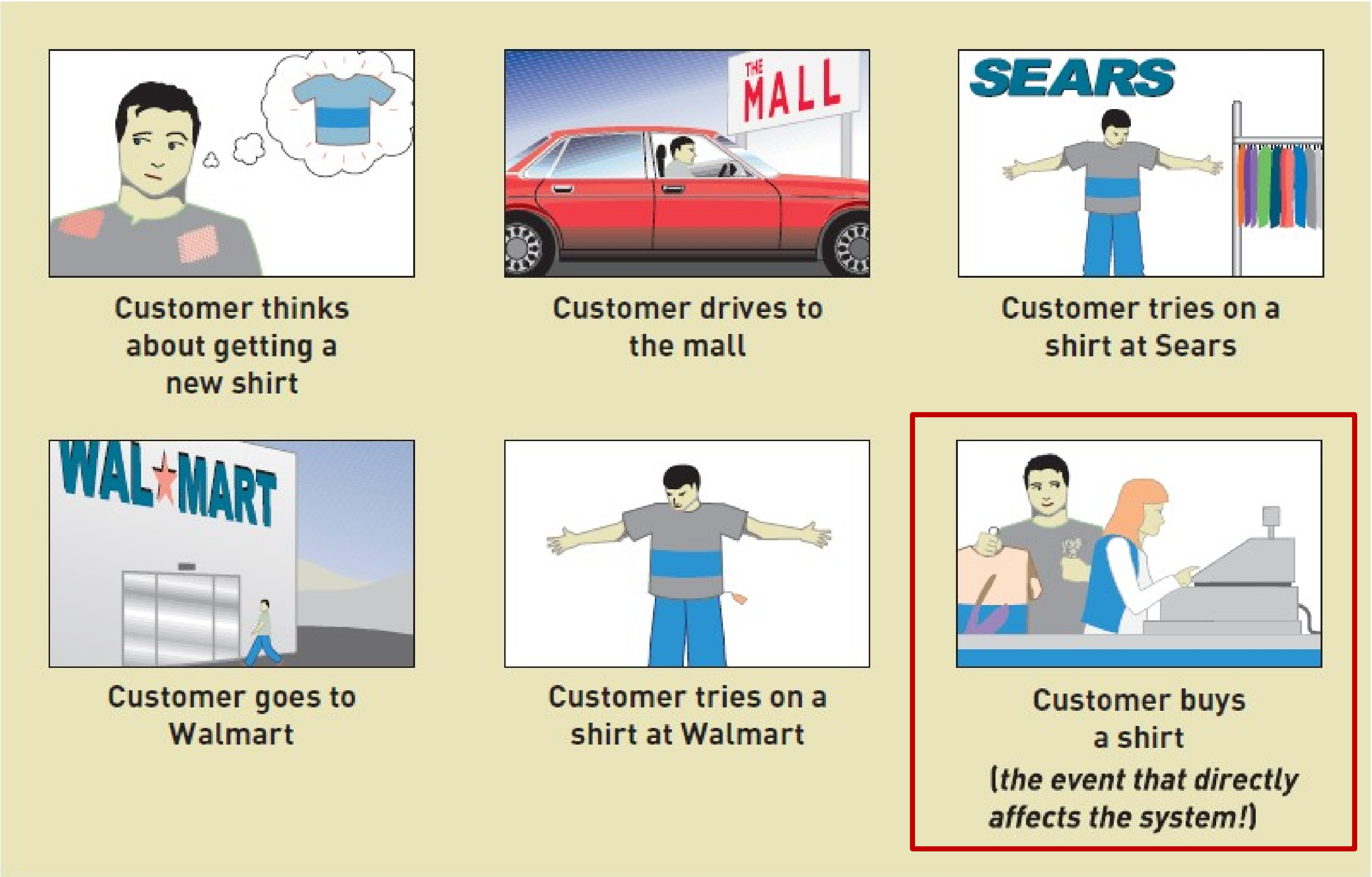
External Event Checklist （外部事件清单）

- External <u>agent or actor</u> wants something resulting in a transaction （外部代理需要导致交易的结果）
  - Customer buys a product （顾客购买产品）
- External <u>agent or actor</u> wants some information （外部代理需要信息）
  - Customer wants to know product details （客户查看产品细节）
- External <u>data changed</u> and needs to be updated （外部数据需要更新）
  - Customer has new address and phone （客户有新地址电话）
- <u>Management</u> wants some information （管理层想要信息）
  - Sales manager wants update on production plans （销售经理想要最新的生产计划）

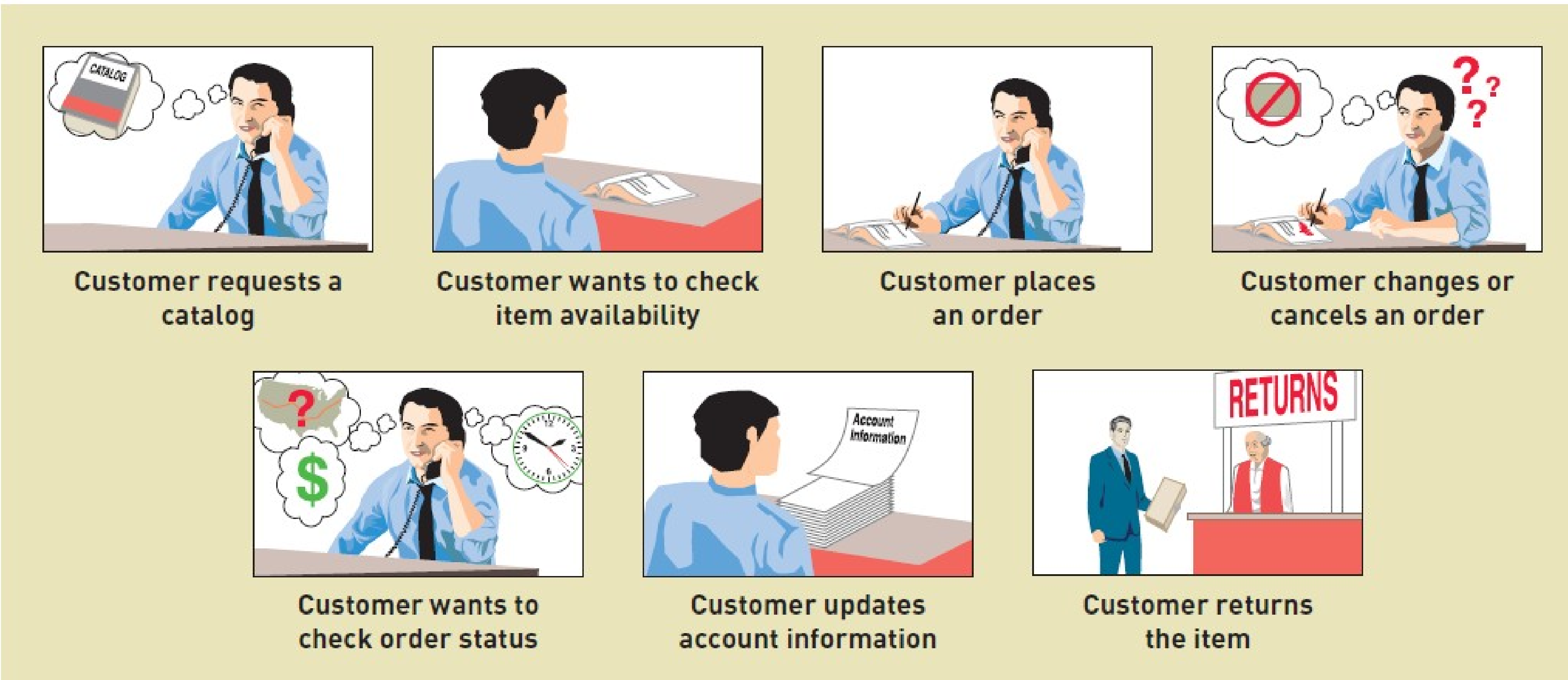
Temporal Event Checklist （时间事件清单）

- Internal outputs needed at points in time （在某时间点进行内部输出）
  - Management reports (summary) （管理报告）
  - Operational reports (detailed transactions) （运营报告）
  - Internal statements and documents (including payroll) （内部报表及文件，如工资单）
- External outputs needed at points of time （在特定时间需要外部输出）
  - Statements, status reports, bills, reminders （报表、报告、账单、提醒）

Finding the actual event that affects the system （查找影响系统的实际事件）

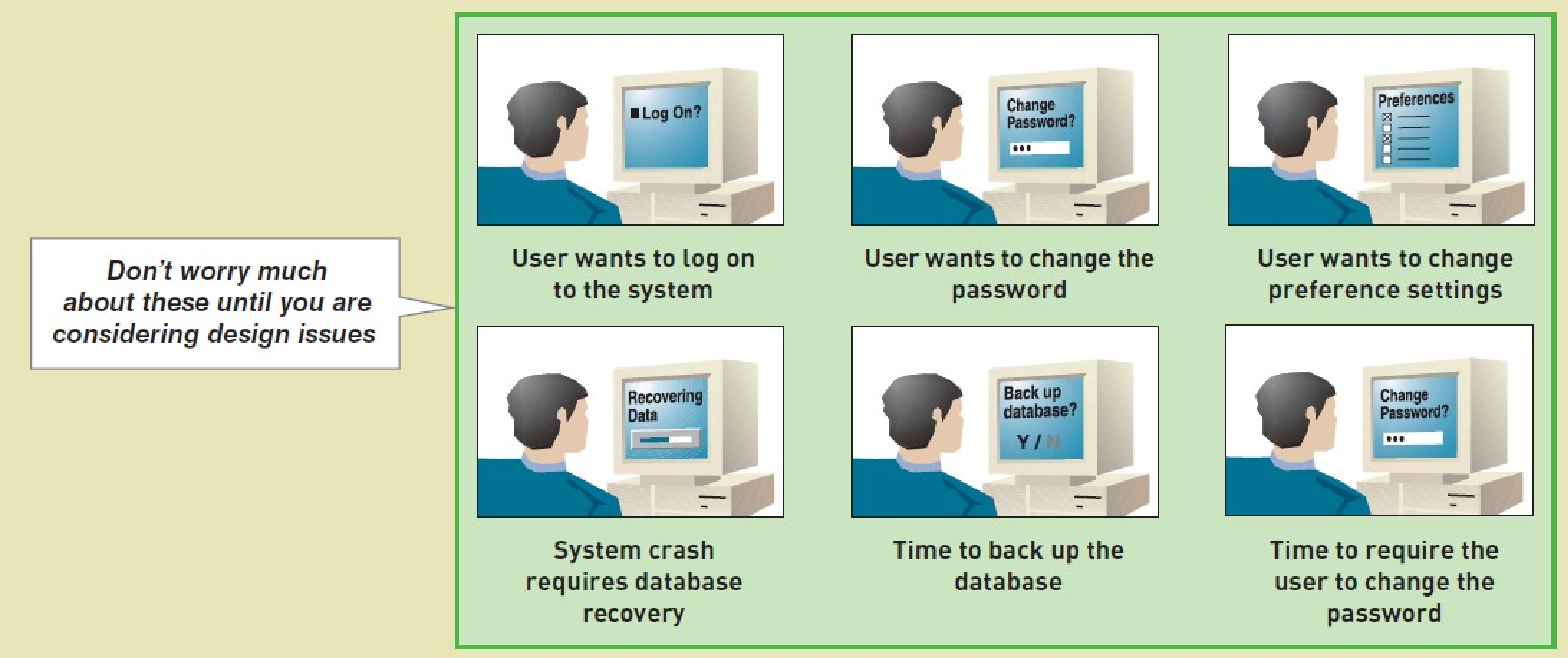


Tracing a sequence of transactions resulting in many events （跟踪导致许多事件的交易序列）



Perfect Technology Assumption （理想技术假设）

- 理想技术假设：只有在最佳条件下系统才需要做出响应，这样的事件才应该在分析阶段考虑进去



Event Decomposition Technique: Specific Steps

1. Consider the external events in the system environment that require a response from the system by using the checklist （考虑系统环境中的外部事件，需要从系统获得响应）
2. For each external event, identify and name the use case that the system requires （对于每个外部事件，设计用例）
3. Consider the temporal events that require a response from the system by using the checklist （考虑需要系统响应的时间事件）
4. For each temporal event, identify and name the use case that the system requires and then establish the point of time that will trigger the use case （对于时间事件，设计用例，建立触发用例的时间点）

Event Decomposition Technique: Specific Steps (continued)

5. Consider the state events （状态事件） that the system might respond to, particularly if it is a real-time system in which devices or internal state changes trigger use cases. （设置状态事件的触发用例）
5. For each state event, identify and name the use case that the system requires and then define the state change. （设计状态事件的状态改变）
5. When events and use cases are defined, check to see if they are required by using the perfect technology assumption. Do not include events that involve such system controls as login, logout, change password, and backup or restore the database, as these are put in later. （检查完美技术假设。不涉及登陆、注销、更改密码、备份数据库等事件，这些事件稍后添加）

Event Decomposition Technique: Benefits

- Events are broader than user goal: Capture temporal and state events （事件比用户目标更广泛：捕获时间和状态事件）
- Help decompose at the right level of analysis: an elementary business process (EBP) （基本业务流程）
- EBP is a fundamental business process performed by one person, in one place, in response to a business event （EBP 是基本的业务流程）
- Uses perfect technology assumption to make sure functions that support the users work are identified and not additional functions for security and system controls （使用完美技术假设来识别用例）

主观题 10 分

设置

List events occurred on the first day when you registered yourself for the university.

注册新生账号
办理学生卡
办理住宿
购买新书
…

正常使用主观题需 2.0 以上版本雨课堂

作答

在学生管理系统中哪些属于外部事件、临时事件和内部事件？

注册新生账号
办理学生卡
办理住宿
购买新书

外部事件

统计学生期末成绩
统计每月跑操时长

时间事件

当学生旷课 **3** 次以上
当学生挂科 **6** 门以上

内部（状态）事件

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Use Cases and CRUD Technique

- CRUD is Create （增） , Read/Report （查） , Update （改） , and Delete (删)
- Technique to validate, refine or cross-check us cases （验证、改进或交叉检查用例的技术）

- 信息系统中最常用、最普遍的功能：
  增、删、改、查、统

Use Cases and CRUD Technique

- For Customer domain class, verify that there are use cases that create, read/report, update, and delete (archive) the domain class

Data entity/domain class	CRUD	Verified use case
Customer	Create	Create customer account
	Read/report	Look up customer Produce customer usage report
	Update	Process account adjustment Update customer account
	Delete	Update customer account (to archive)

CRUD Technique Steps

1. Identify all the data entities or domain classes involved in the new system. (more in Chapter 4) （确定所有包含在新系统中的数据实体或域类）
2. For each type of data (data entity or domain class), verify that a use case has been identified that creates a new instance, updates existing instances, reads or reports values of instances, and deletes (archives) an instance. （对于每种数据来说，要核实已确定的用例，包括创建一个新实例、更新现有的实例、读取或报告实例的价值以及删除（存储）实例）

CRUD Technique Steps

3. If a needed use case has been overlooked, add a new use case and then identify the stakeholders. （如果一个必须的用例被忽视了，那么需要添加一个新用例然后确定利益相关者）
3. With integrated applications, make sure it is clear which application is responsible for adding and maintaining the data and which system merely uses the data. （通过已整合的应用来保证能清晰地指出哪个应用是负责添加和维护数据的，而哪个系统仅仅是使用数据的）

CRUD Technique
Use Case vs. Domain Class Table

- To summarize CRUD analysis results, create a matrix of use cases and domain classes indicating which use case C, R, U, or D a domain class

Use case vs. entity/domain class	Customer	Account	Sale	Adjustment
Create customer account	C	C		
Look up customer	R	R		
Produce customer usage report	R	R	R	
Process account adjustment	R	U	R	C
Update customer account	UD (archive)	UD (archive)		

Use Cases and Brief Use Case Descriptions

- Brief use case description is often a one sentence description showing the main steps in a use case

Use case	Brief use case description
Create customer account	User/actor enters new customer account data, and the system assigns account number, creates a customer record, and creates an account record.
Look up customer	User/actor enters customer account number, and the system retrieves and displays customer and account data.
Process account adjustment	User/actor enters order number, and the system retrieves customer and order data; actor enters adjustment amount, and the system creates a transaction record for the adjustment.

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RMO CSMS Project Use Cases

CSMS 销售子系统

CSMS sales subsystem	
Use cases	**Users/actors**
Search for item	Customer, customer service representative, store sales representative
View product comments and ratings	Customer, customer service representative, store sales representative
View accessory combinations	Customer, customer service representative, store sales representative
Fill shopping cart	Customer
Empty shopping cart	Customer
Check out shopping cart	Customer
Fill reserve cart	Customer
Empty reserve cart	Customer
Convert reserve cart	Customer
Create phone sale	Customer service representative
Create store sale	Store sales representative

RMO CSMS Project Use Cases

CSMS 订单执行子系统

CSMS order fulfillment subsystem	
Use cases	**Users/actors**
Ship items	Shipping
Manage shippers	Shipping
Create backorder	Shipping
Create item return	Shipping, customer
Look up order status	Shipping, customer, management
Track shipment	Shipping, customer, marketing
Rate and comment on product	Customer
Provide suggestion	Customer
Review suggestions	Management
Ship items	Shipping
Manage shippers	Shipping

RMO CSMS Project Use Cases

CSMS 顾客账户子系统

CSMS Customer account subsystem	
Use cases	**Users/actors**
Create/update customer account	Customer, customer service representative, store sales representative
Process account adjustment	Management
Send message	Customer
Browse messages	Customer
Request friend linkup	Customer
Reply to linkup request	Customer
Send/receive points	Customer
View “mountain bucks”	Customer
Transfer “mountain bucks”	Customer

RMO CSMS Project Use Cases

CSMS 市场营销子系统

CSMS marketing subsystem	
Use cases	**Users/actors**
Add/update product information	Merchandising, marketing
Add/update promotion	Marketing
Add/update accessory package	Merchandising
Add/update business partner link	Marketing

CSMS reporting subsystem	
Use cases	**Users/actors**
Produce daily transaction summary report	Management
Produce sales history report	Management, marketing
Produce sales trends report	Marketing
Produce customer usage report	Marketing
Produce shipment history report	Management, shipping
Produce promotion impact report	Marketing
Produce business partner activity report	Management, marketing

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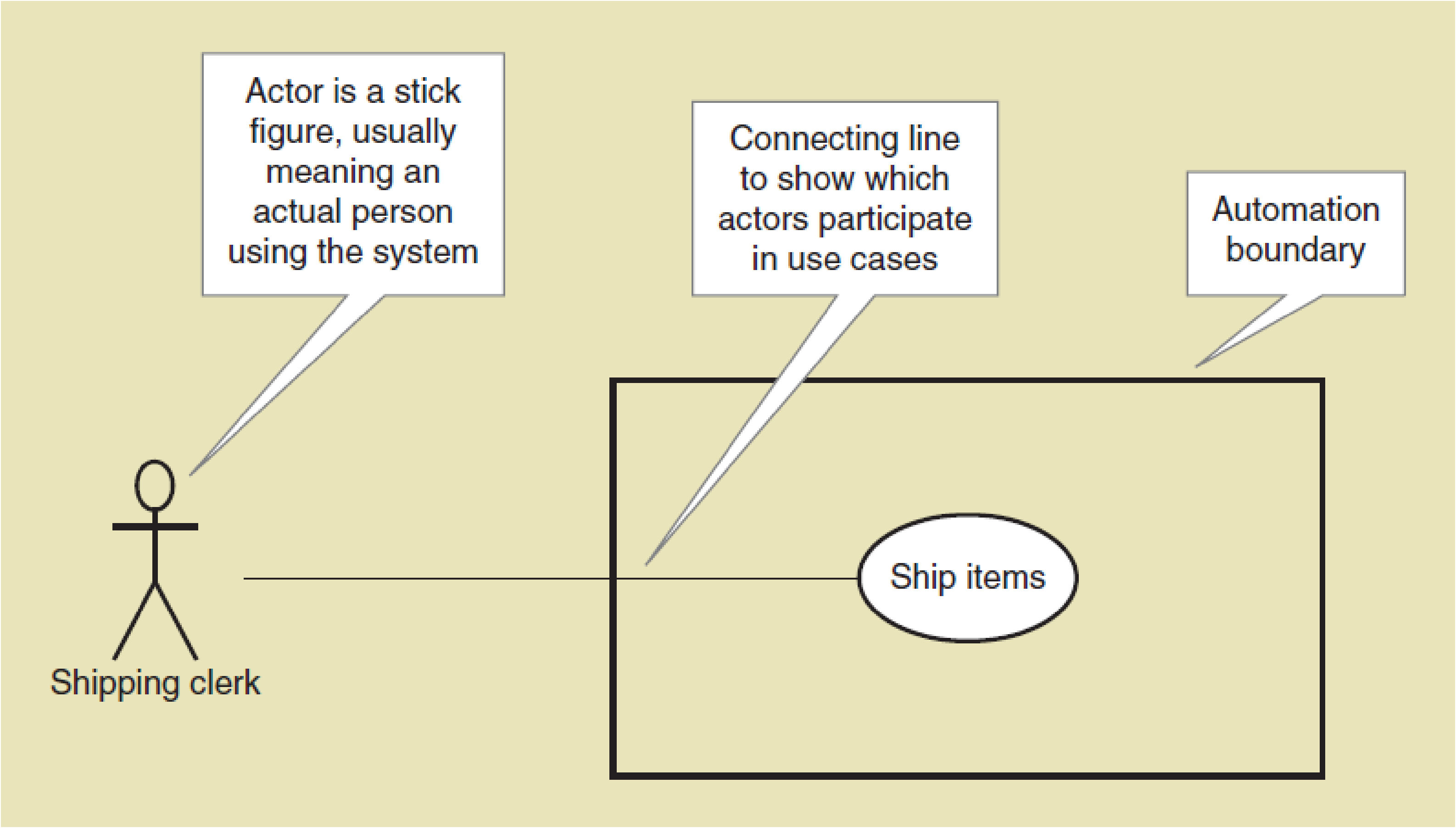
Use Case Diagrams

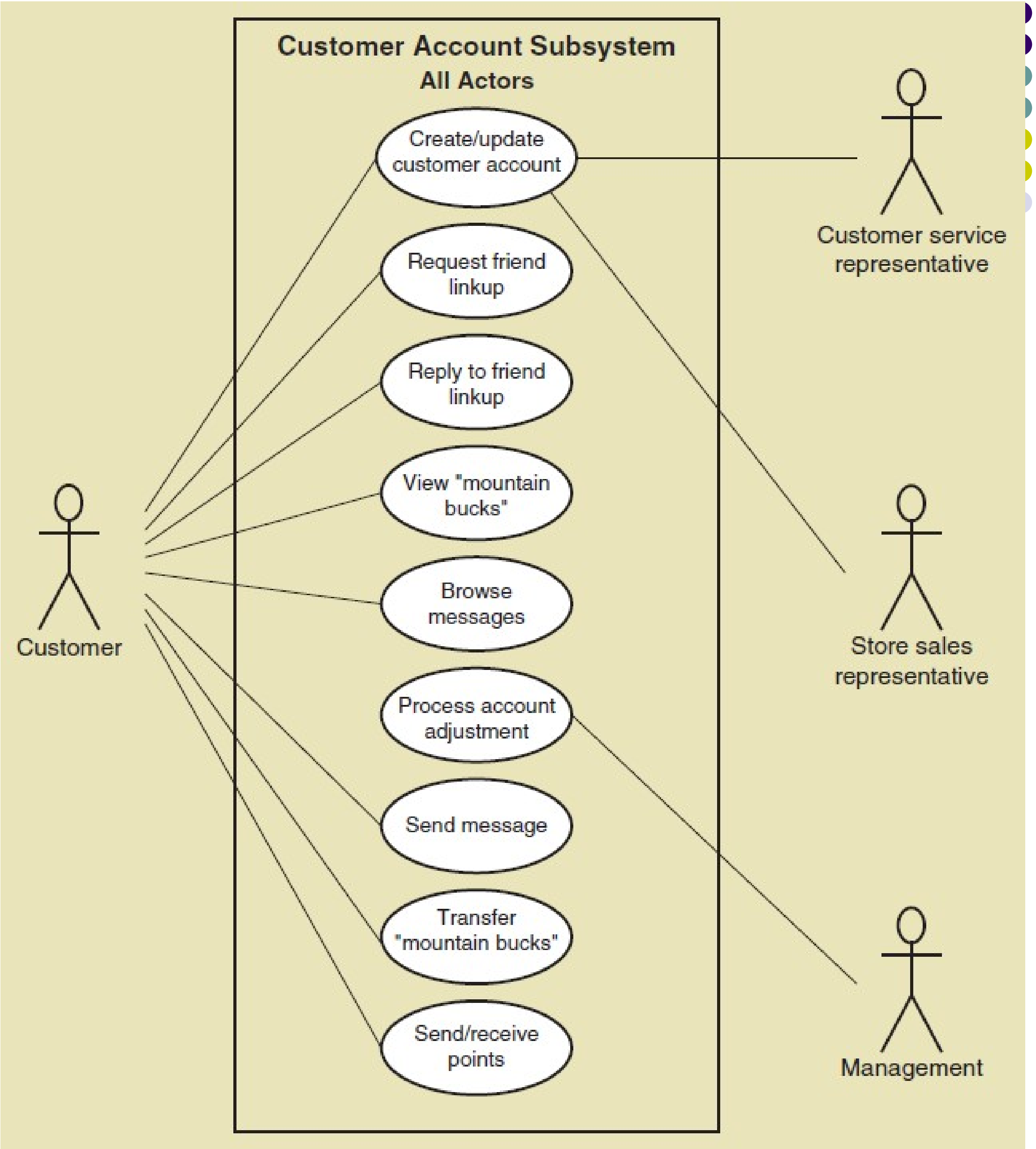
- Use case diagram （用例图）— a UML model used to graphically show uses cases and their relationships to actors （一种 UML 模型，用于图形化地显示用例及其与参与者的关系）
- Recall UML is Unified Modeling Language （统一建模语言）, the standard for diagrams and terminology for developing information systems
- Automation boundary （自动化边界）— the boundary between the computerized portion of the application and the users who operate the application （应用程序的计算机化部分和操作应用程序的用户之间的边界）

Use Case Diagrams Symbols



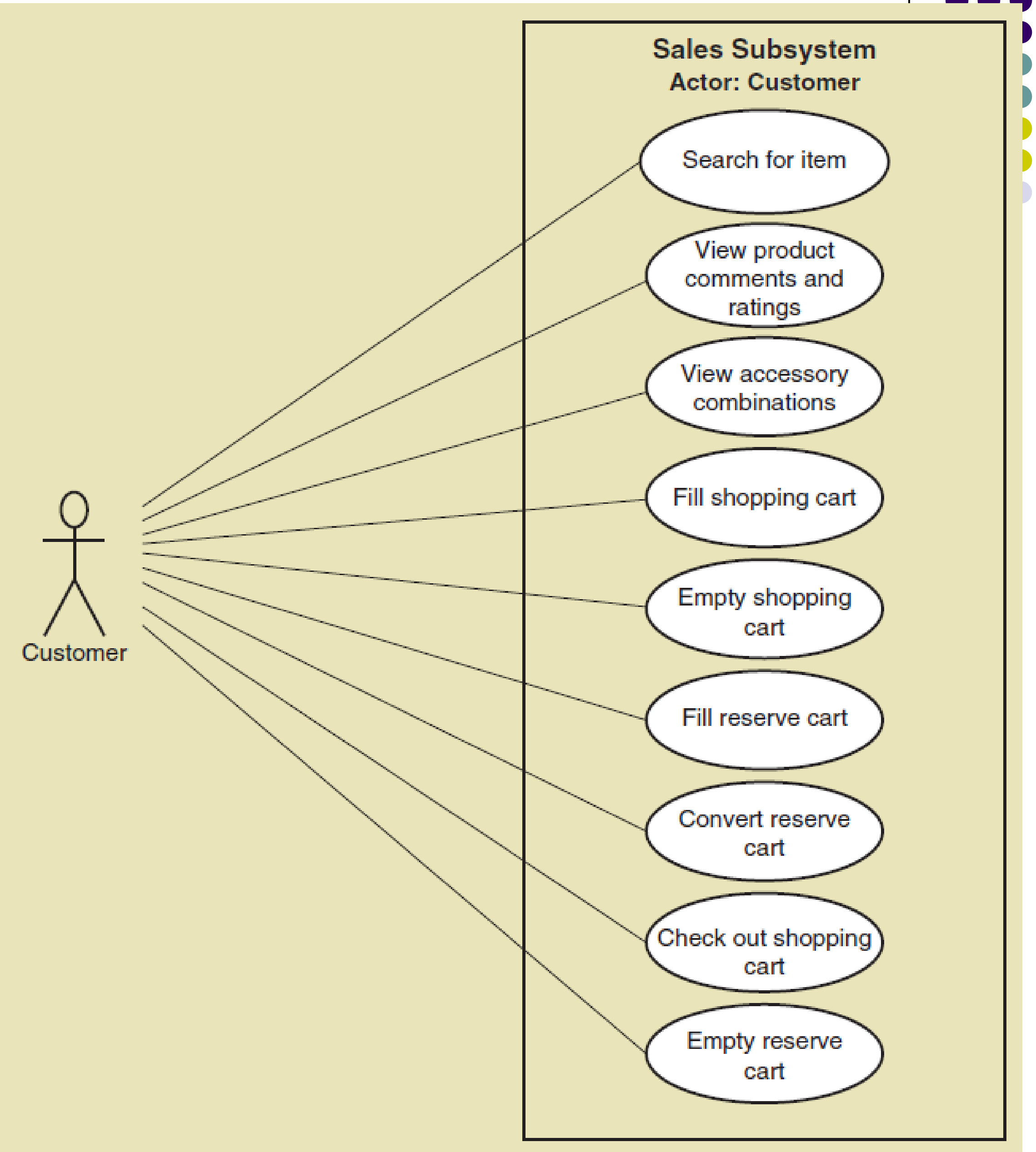
Use Case Diagrams

Draw for each subsystem

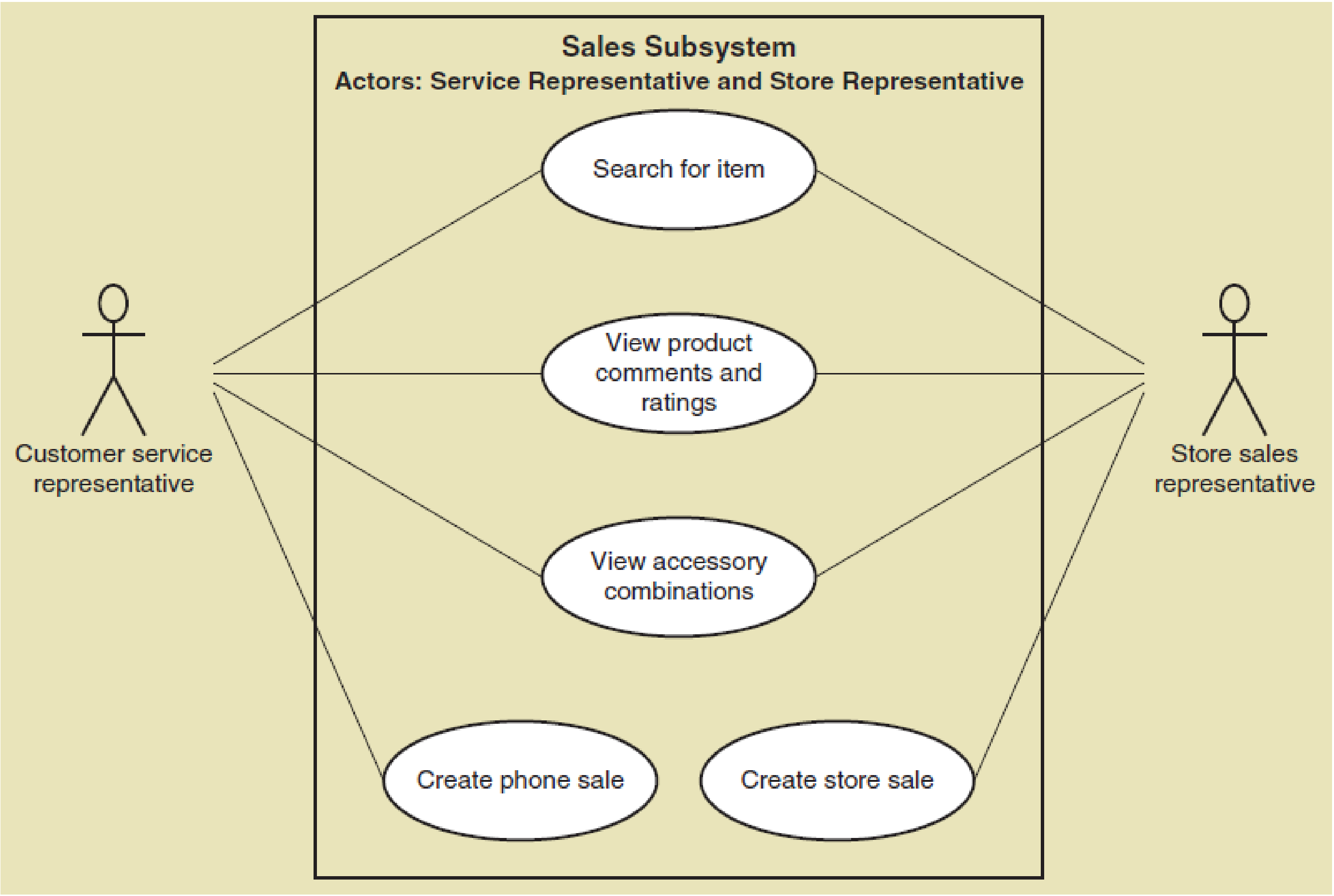


Use Case Diagrams

Draw for actor, such as customer



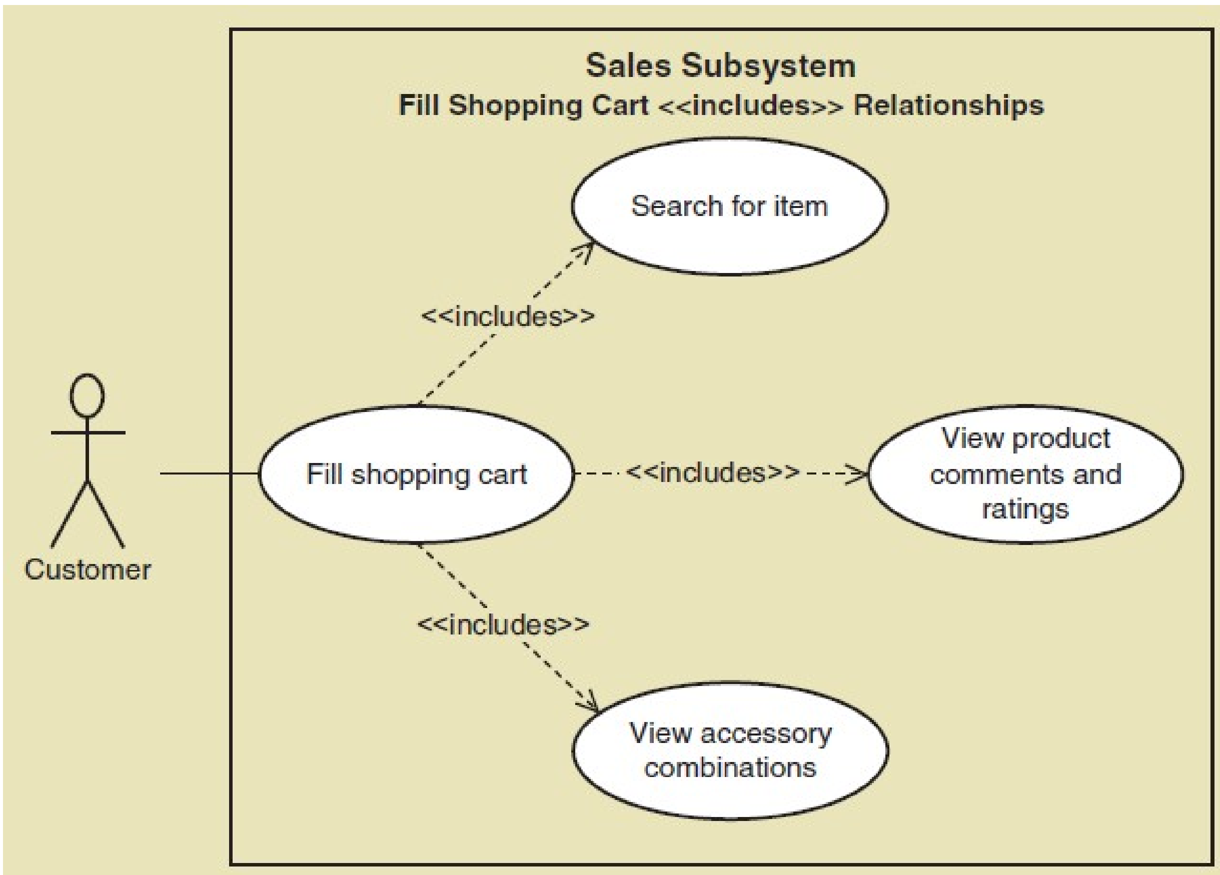
Use Case Diagrams
Draw for internal RMO actors



Use Case Diagrams

The <<Includes>> relationship

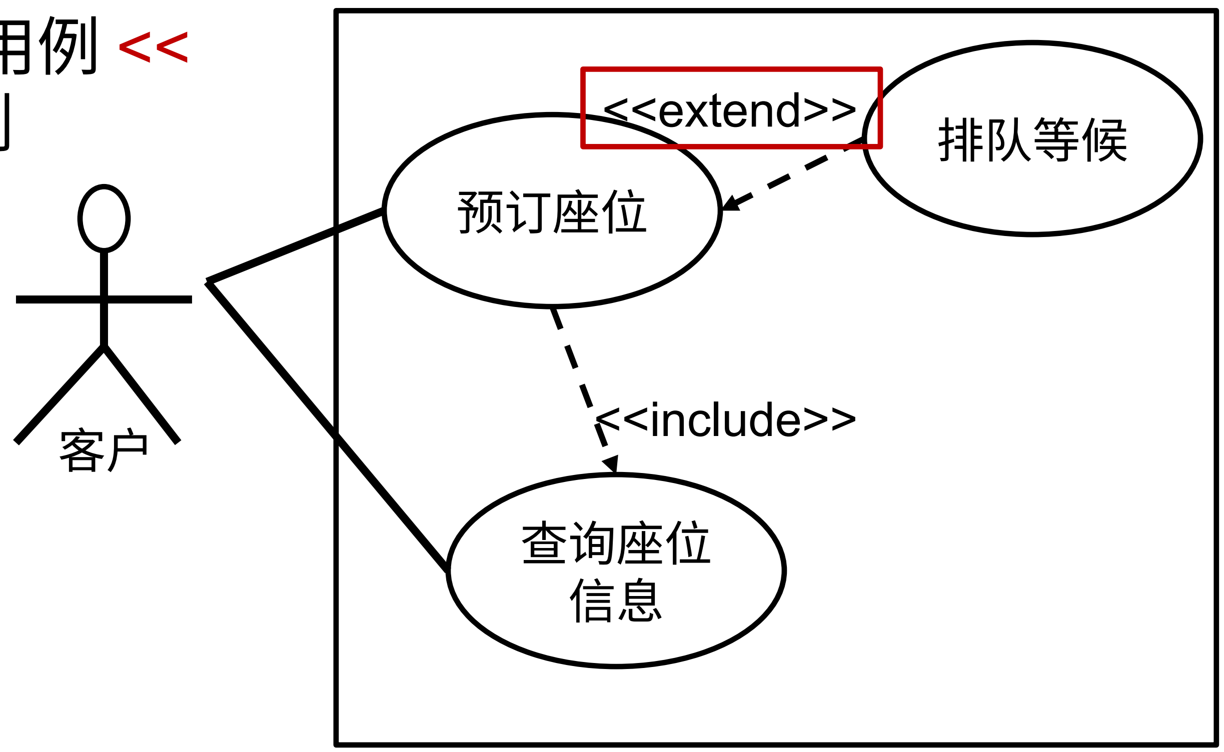
- A relationship between use cases where one use case is stereotypically included within the other use case— like a called subroutine. Arrow points to subroutine



Use Case Diagrams

The <<Extend>> relationship

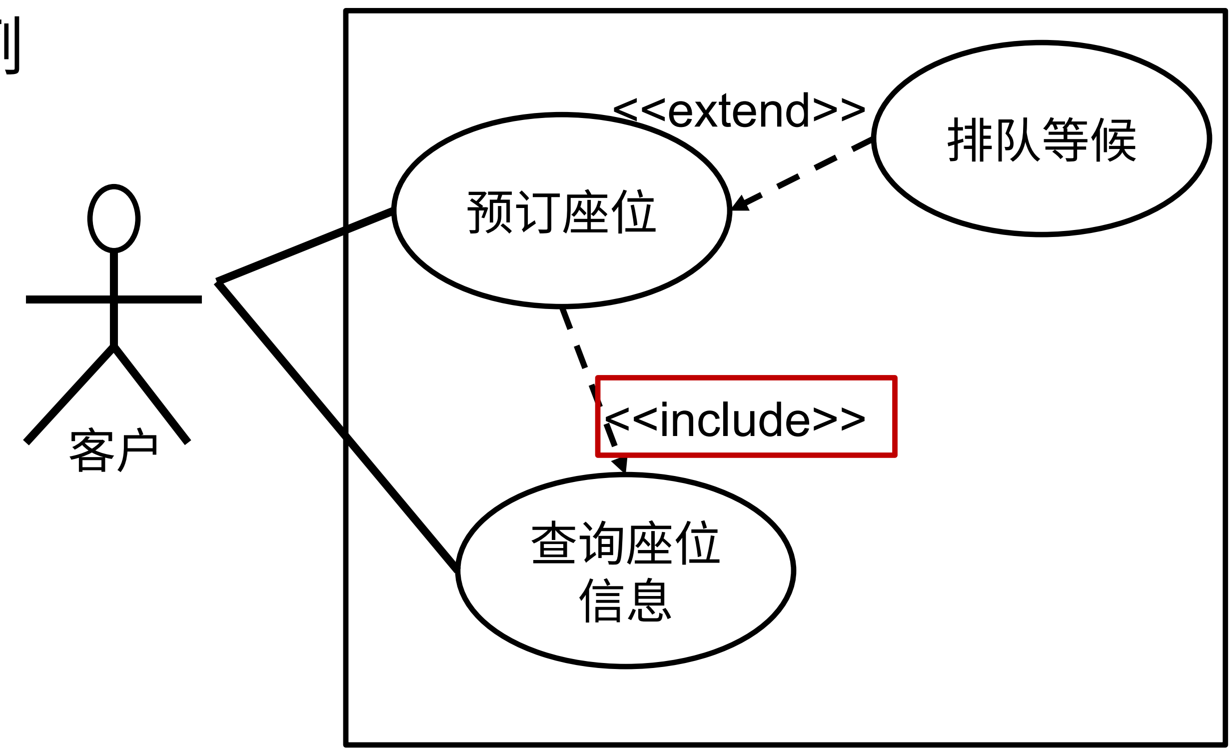
- 当执行 A 用例时，有些情况下，会执行 B 用例，有些情况下不会执行 B 用例
- 这种情况就是 B 用例 <<扩展>> 了 A 用例



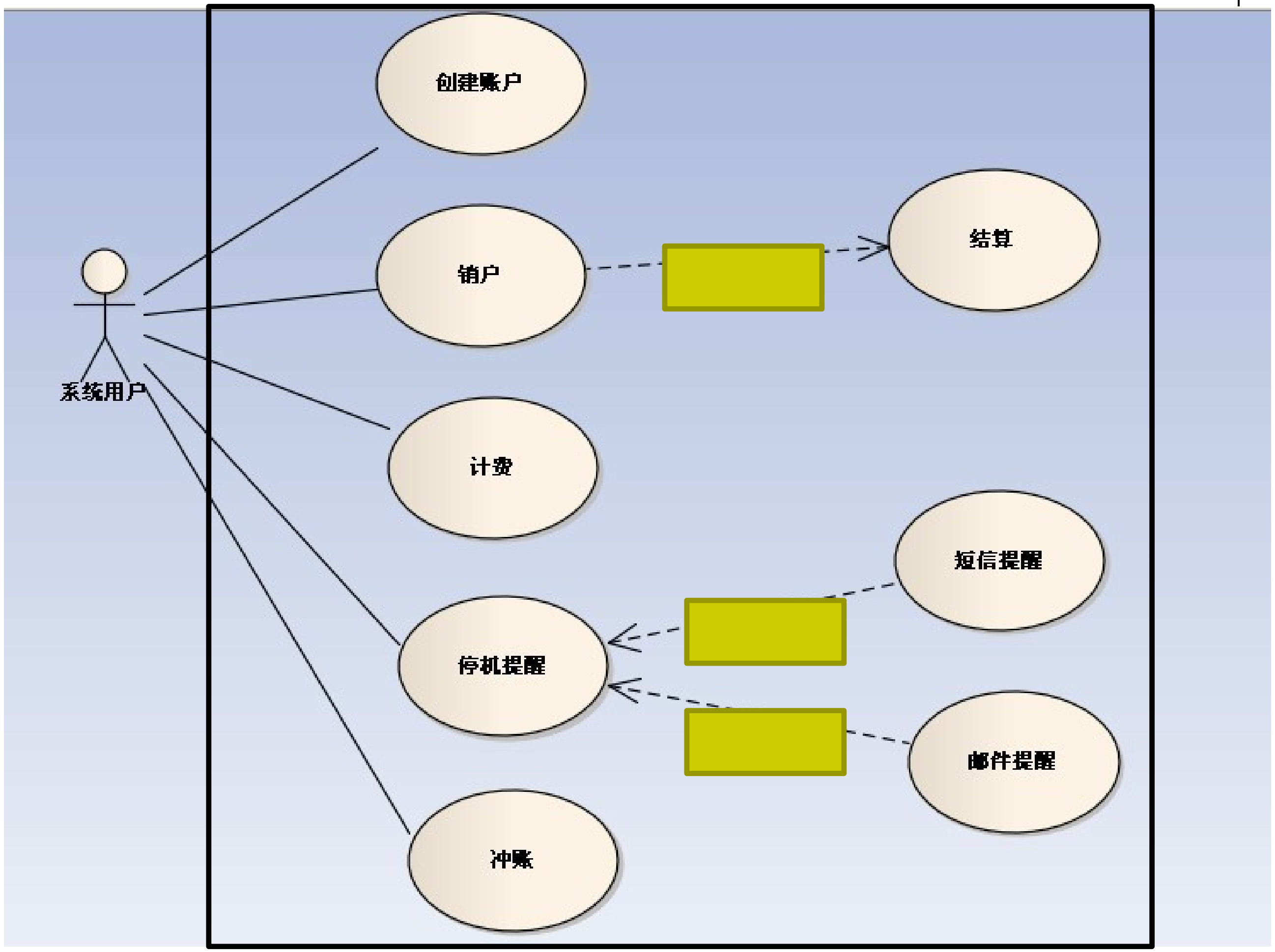
Use Case Diagrams

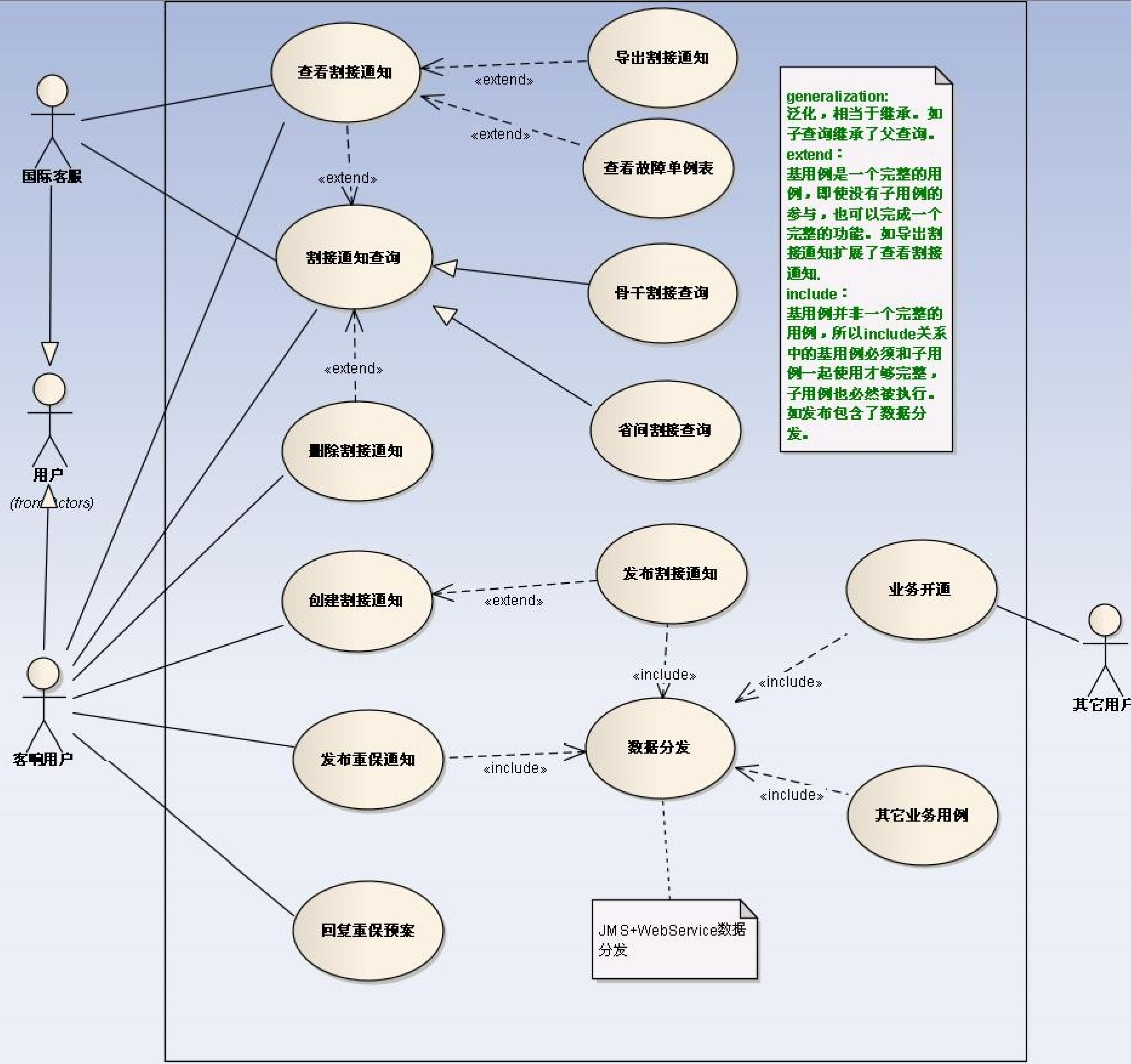
The <<Extend>> relationship

- 当执行 A 用例时，<u>必须执行 B 用例</u>才完整
- 这种情况就是 A 用例 <<包含>> 了 A 用例



An Use Case Diagram Example




国际客服
用户
(from Actors)
客响用户
其它用户
查看割接通知
导出割接通知
«extend»
查看故障单例表
«extend»
«extend»
割接通知查询
骨干割接查询
省间割接查询
«extend»
删除割接通知
创建割接通知
发布割接通知
«extend»
业务开通
«include»
«include»
数据分发
发布重保通知
«include»
其它业务用例
«include»
回复重保预案
JMS+WebService数据分发
generalization:
泛化，相当于继承。如子查询继承了父查询。
extend：
基用例是一个完整的用例，即使没有子用例的参与，也可以完成一个完整的功能。如导出割接通知扩展了查看割接通知.
include：
基用例并非一个完整的用例，所以include关系中的基用例必须和子用例一起使用才够完整，子用例也必然被执行。如发布包含了数据分发。

Use Case Diagrams: Steps

1. Identify all the stakeholders and users who would benefit by seeing a use case diagram （确定相关人员）
2. Determine what each stakeholder or user needs to review in a use case diagram: each subsystem, for each type of user, for use cases that are of interest （确定相关人员想在用例图中看到什么）
3. For each potential communication need, select the use cases and actors to show and draw the use case diagram. There are many software packages that can be used to draw use case diagrams （选择要显示的用例和参与者，并绘制用例图）
4. Carefully name each use case diagram （命名每个用例图） and then note how and when the diagram should be used to review use cases with stakeholders and users

Summary

- This chapter is the first of three that focuses on modeling functional requirements as a part of systems analysis （本章是将功能需求建模作为系统分析一部分的三章中的第一章）
- Use cases are the functions identified, the activities the system carries out usually in response to a user request （用例是识别的功能，系统通常为响应用户请求而执行的活动）
- Two techniques for identifying use cases are the user goal technique and the event decomposition technique 识别用例的两种技术是用户目标技术和事件分解技术

Summary

- The user goal technique begins by identifying end users called actors and asking what specific goals they have when interacting with the system （用户目标技术首先确定被称为参与者的最终用户，并询问他们在与系统交互时具有哪些特定的目标）
- The event decomposition technique begins by identifying events that occur that require the system to respond. （事件分解技术首先识别需要系统响应的事件）

Summary

- Three types of events include external, temporal, and state events （三种类型的事件包括外部事件、时间事件和状态事件）
- Brief use case descriptions are written for use cases （简短的用例描述是为用例编写的）
- The CRUD technique is used to validate and refine the use cases identified （CRUD 技术用于验证和细化已识别的用例）
- The use case diagram is the UML diagram used to show the use cases and the actors （用例图是用来显示用例和参与者的 UML 图）

Summary

- The use case diagram shows the actors, the automation boundary, the uses cases that involve each actor, and the <<includes>> <<extend>>relationship. （用例图显示了参与者、自动化边界、涉及每个参与者的用例，以及 << 包括 >> << 扩展 >> 关系。）
- A variety of use case diagrams are draw depending on the presentation needs of the analysis （根据分析的表示需要绘制各种用例图）